The study used specimens made from cement (CEM I 52.5N), natural river sand, silica fume (SF), and LM. A polycarboxylate-based superplasticizer (SP) was used to compensate for the decrease in fresh mortar flowability caused by LM. The chemical composition of the binding materials, which was obtained with X-ray fluorescence analysis (XRF; T8 Tiger, USA), is shown in Table 1. Moreover, LM, SF, and cement have a mean particle size of 21.7 μm, 3.98 μm, and 10 μm. Additionally, the LM has a density of 2.7 g/cm$^3$

Table 1. Chemical composition of raw materials (wt. %).

| | CaO | $SiO_2$ | $Al_2O_3$ | $Fe_2O_3$ | MgO | $SO_3$ | $Na_2O$ | $K_2O$ | Others | LOI |
|---|---|---|---|---|---|---|---|---|---|---|
| Cement | 64.1 | 17.8 | 4.4 | 3.7 | 3.3 | 3.9 | 0.5 | 1.4 | 1.0 | 3.1 |
| SF | 0.1 | 97.3 | 0.1 | 0.2 | 0.4 | 0.9 | - | - | 1.0 | 2.0 |
| LM | 90.4 | 0.8 | 0.5 | 0.9 | 1.4 | 1.3 | 2.5 | 0.1 | 2.1 | 43.1 |

**2.1 Sample preparation**

The specimens were prepared using ASTM C305 (2020) with a water-to-binder ratio of 0.6 and a sand-to-binder ratio of 1 as shown in Table 2. The SP used was 2.5%–3.0%, the weight of the binder, to maintain the same flowability. After mixing the materials following the standard, the fresh paste was then poured into a 0.05 × 0.05 × 0.05 m$^3$ mold, which was sealed with a plastic bag and left at room temperature for 24 h. After 24 h, the samples were demolded and cured in steam curing for 48 h at (90 °C).

Table 2. Mix design for the mortar specimens (kg/m$^3$).

| Mixture ID | Cement | Silica fume | Lime mud | Sand | Water | SP |
|---|---|---|---|---|---|---|
| REF | | | 0.0 | 923.1 | | 4.6 |
| LM50 | 738.5 | 184.6 | 461.5 | 461.5 | 553.8 | 13.8 |
| LM75 | | | 692.3 | 230.8 | | 18.5 |
| LM100 | | | 923.1 | 0.0 | | 27.7 |

**2.2 Testing methods**

The compressive strength tests were performed using ASTM C109/109M-16a (2016) with a loading rate of 0.5 mm/min. The average of the three replicates was then used as a representative test result. For the quantification of hydration products thermogravimetric analysis (TGA; Q500, TA Instruments USA) was used. Mercury intrusion porosimetry (MIP) measurements were performed with Autopore VI 9620 (Micromeritics Instrument Corp., USA) at pressures ranging from 0.2 to 413.7 MPa to investigate the microstructure of the mortar samples.

**3. Result and Discussion**

**3.1 Mechanical strength**

The compressive strength of different specimens is shown in Figure 1. Compared to the reference sample, the LM50, LM75, and LM100 specimens exhibit varying degrees of increased compressive strength, with the LM100 sample showing the largest increase of 39.5%. This improvement is mainly due to the micro-aggregate effect of LM and SF, which fills the pores and improves the pore structure. More detailed information on this topic can be found in Section 3.3. Previous studies have also demonstrated that LM creates nucleation sites that enhance strength, contributing to the increase in strength observed (Goldman and Bentur 1993; Singh et al. 2015).